

RANDI YULIO FAJRI

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PROFESSIONAL SUMMARY

Product-oriented Backend Engineer / Fullstack Developer with extensive experience architecting scalable backend services, event-driven distributed systems, and high-throughput microservices using Go (Golang), Node.js, and Python. Proven track record in designing resilient communication patterns with RabbitMQ, Kafka, and gRPC, alongside rigorous SQL query optimization, database performance tuning (PostgreSQL/MySQL), and Redis caching. Experienced in developing intelligent backend applications by building ML/AI pipelines and integrating AI/LLM APIs with LangChain and Vector Databases. Adept at translating complex business requirements into high-performance, fault-tolerant technical architectures.

TECHNICAL SKILLS

Backend & Languages: Go (Golang), Node.js, Python, TypeScript/JavaScript, PHP (Laravel)

Frameworks & APIs: Fastify, Express.js, FastAPI, Django, Gin, RESTful API Design, gRPC (Protobuf)

Databases & Caching: PostgreSQL, MySQL, Redis, SQL Query Optimization, Indexing, Database Performance Tuning

Distributed Systems & Queue: Microservices, Concurrency (Go Goroutines/Channels), Kafka, RabbitMQ, Event-Driven Architecture

AI, ML & LLM: LLM API Integration (OpenAI, Anthropic, Gemini), LangChain, Vector Databases (pgvector, Pinecone), Biometrics & ML Pipelines

DevOps & Observability: Docker, Kubernetes, Git, Prometheus, Grafana, Loki, Alertmanager

Frontend: React.js, Next.js, Vue.js, Nuxt.js, Tailwind CSS

WORK EXPERIENCE

Fullstack Web Developer | Jan. 2026 – Present

PT. Trisan Esa Cipta — Palembang, Indonesia (Full-time)

- Architected scalable backend microservices using **Fastify (Node.js)** and **Go** to power enterprise HR management and attendance tracking platforms.
- Engineered an ML/AI verification pipeline combining **Python (FastAPI)** face recognition models with **Vector Databases** for sub-second biometric embedding matching.
- Integrated **LLM APIs** via **LangChain** to build an automated policy assistant, allowing employees to query internal HR documentation via natural language.
- Designed high-throughput RESTful endpoints and implemented **Redis** for distributed caching and

session management, reducing API latency for concurrent clock-in requests.

- Conducted database performance tuning and SQL query optimization on MySQL, leveraging indexes and query plan analysis (EXPLAIN ANALYZE / EXPLAIN) to maintain rapid execution times.
- Containerized multi-tier backend services using **Docker** and established structured Git workflows for reliable cross-platform CI/CD deployments.

Distributed Systems Engineer | Sept. 2024 – Dec. 2025

OXY Creative — Jakarta Selatan, Indonesia (Full-time)

- Developed highly scalable distributed backend architectures using **Go** and **Node.js (Express.js)** to handle heavy inventory transactional flows.
- Implemented inter-service communication using **gRPC** with Protocol Buffers, cutting inter-service latency and network overhead across internal microservices.
- Orchestrated event streaming and asynchronous queue processing using **RabbitMQ** as a message broker to decouple service communications and real-time alerts.
- Managed complex relational schemas on **MySQL and PostgreSQL**; resolved deadlocks and optimized query execution paths for high-frequency inventory write operations.
- Leveraged Go's native **concurrency primitives (goroutines, channels, and worker pools)** to parallelize bulk inventory reconciliation workflows.
- Collaborated with DevOps to deploy microservices on **Kubernetes**, configuring comprehensive observability through Prometheus, Grafana, Loki, and Grafana Alloy.

Fullstack Developer | Mar. 2024 – Jun. 2025

PT. Immobi Solusi Prima — Jakarta Selatan, Indonesia (Contract / Full-time)

- Delivered enterprise-grade fullstack web services for Telkomsel's centralized dashboard platform (AVP), integrating React, Next.js, and backend authentication microservices.
- Built performant backend aggregation APIs connecting diverse data sources (PowerBI, Grafana, and relational datastores) with single sign-on (SSO) and RBAC security.
- Employed a product-oriented mindset to translate stakeholder and enterprise client requirements into resilient backend pipelines capable of exporting large-scale datasets.

Software Engineer | Feb. 2022 – Mar. 2024

OXY Creative — Jakarta Selatan, Indonesia (Full-time)

- Architected an event-driven microservices architecture processing millions of onboarding records across banking and rental identity validation platforms.
- Engineered asynchronous pipeline workflows with **RabbitMQ**, guaranteeing fault-tolerant messaging, retry policies, and dead-letter queues.
- Integrated national identity APIs (Dukcapil), credit assessment systems, and facial biometric verification pipelines using **Python (Django / FastAPI)**.
- Optimized high-volume transactional queries on **PostgreSQL**, implementing connection pooling, composite indexing, and database tuning to sustain heavy write loads.

Fullstack Developer | Feb. 2021 – Feb. 2022

PT. Hedo Global Technology — Palembang, Indonesia (Full-time)

- Developed and maintained core warehouse management APIs, processing inventory movements, damaged goods returns, and automated e-commerce stock syncing.
- Refactored legacy SQL queries and backend services, improving database throughput and minimizing data inconsistency across concurrent sales channels.

EDUCATION

Associate Degree in Informatics Management | Aug. 2017 – Oct. 2020

State Polytechnic of Sriwijaya (Politeknik Negeri Sriwijaya) — Palembang, Indonesia
Graduated